

The thing that makes tail events easy to underappreciate is how easy it is to underestimate how things compound. How, for example, 9/11 prompted the Federal Reserve to cut interest rates, which helped drive the housing bubble, which led to the financial crisis, which led to a poor jobs market, which led tens of millions to seek a college education, which led to \$1.6 trillion in student loans with a 10.8% default rate. It's not intuitive to link 19 hijackers to the current weight of student loans, but that's what happens in a world driven by a few outlier tail events.

The majority of what's happening at any given moment in the global economy can be tied back to a handful of past events that were nearly impossible to predict.

The most common plot of economic history is the role of surprises. The reason surprises occur is not because our models are wrong or our intelligence is low. It's because the odds that Adolf Hitler's parents argued on the evening nine months before he was born were the same as them conceiving a child. Technology is hard to predict because Bill Gates may have died from polio if Jonas Salk got cranky and gave up on his quest to find a vaccine. The reason we couldn't predict the student loan growth is because an airport security guard may have confiscated a hijacker's knife on 9/11. That's all there is to it.

The problem is that we often use events like the Great Depression and World War II to guide our views of things like worst-case scenarios when thinking about future investment returns. But those record-setting events had no precedent when they occurred. So the forecaster who assumes the worst (and best) events of the past will match the worst (and best) events of the future is not following history; they're accidentally assuming that the history of unprecedented events doesn't apply to the future.

Nassim Taleb writes in his book *Fooled By Randomness*:

In Pharaonic Egypt ... scribes tracked the high-water mark of the Nile and used it as an estimate for a future worst-case scenario. The same can be seen in the Fukushima nuclear reactor, which experienced a catastrophic failure in 2011 when a tsunami struck. It had been built to withstand the worst past historical earthquake, with the builders not imagining much worse—and not thinking that the worst past event had to be a surprise, as it had no precedent.